International Factor Mobility and Foreign Investment

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International Factor Mobility

In economics, the `factors of production` are primarily **Labor** (workers) and **Capital** (machinery, factories, financial investment).

- **Traditional Trade Theory (e.g., Heckscher-Ohlin)** assumes that factors are `perfectly mobile domestically` but `completely immobile internationally`. Countries trade goods to **indirectly** "exchange" their abundant factors.
- **International Factor Mobility** relaxes this assumption, allowing labor `(migration)` and capital `(Foreign Direct Investment - FDI`, or `portfolio investment`) to move directly across borders.

The Core Theoretical Debate: Substitutes vs. Complements

Does `factor mobility` replace `international trade`, or does it increase it? Economists have two main views:

A. The Substitution Hypothesis (Mundell, 1957)

- **Logic:** Based on the Heckscher-Ohlin model. If Country A is capital-abundant and Country B is labor-abundant, Country A will export capital-intensive goods to Country B.
- **If factors can move:** Capital will simply flow from Country A to Country B to chase higher returns, and labor might flow from B to A.
- **Result:** Factor mobility **reduces or eliminates** the need for trade in goods. They are **substitutes**.

B. The Complementarity Hypothesis (Markusen, 1983, 1984; Helpman, 1984)

- **Logic:** In the real world, trade and factor mobility often feed each other, especially through **Multinational Corporations (MNCs)**.
- **Vertical FDI:** A US tech firm (capital/technology) builds a factory in Vietnam (labor). This capital movement `increases` trade, because the US must `export intermediate components` to Vietnam, and Vietnam exports the `finished goods` back to the US.
- **Result:** Factor mobility and trade are **complements**.

Economic Effects of Labor Mobility (Migration)

When workers move from a low-wage country (`Source`) to a high-wage country (`Host`), the global allocation of labor becomes more efficient, but there are distinct winners and losers.

Host Country (Immigration)

- **Labor (Native Workers): Lose**. An increase in labor supply drives down the equilibrium wage for native workers (especially those with similar skills to the immigrants).
- **Capital Owners / Firms: Win**. Cheaper labor increases profit margins and the return on capital.
- **Net National Welfare: Increases**. The gains to capital owners and consumers (lower prices for services/goods) outweigh the losses to native workers. There is a "migration dividend."

Source Country (Emigration)

- **Labor (Those who stay): Win**. With fewer workers, the marginal product of labor rises, leading to higher wages for those remaining.
- **Capital Owners: Lose**. Higher wages reduce profit margins and the return on domestic capital.
- **Net National Welfare: Increases** (due to higher wages for stayers + **remittances** sent back by migrants).
- **Caveat (Brain Drain):** If the emigrants are highly skilled (doctors, engineers), the source country loses its most valuable human capital, which can severely harm long-term development.

Economic Effects of Capital Mobility (FDI)

When `capital` flows from a capital-abundant country (`Source`) to a capital-scarce country (`Host`), it seeks higher returns.

Host Country (Capital Inflow)

- **Labor: Wins**. More capital per worker increases labor productivity, driving up wages.
- **Domestic Capital Owners: Lose**. Increased competition from foreign capital drives down the domestic return on capital (interest rates/profits fall).
- **Net National Welfare: Increases**. The wage gains for labor outweigh the losses to domestic capital owners.

Source Country (Capital Outflow)

- **Capital Owners: Win**. They earn higher returns on their investments abroad than they would at home.
- **Labor: Loses**. As capital leaves, the capital-to-labor ratio falls, reducing labor productivity and pushing domestic wages down.
- **Net National Welfare: Increases**. The higher returns earned by capital owners outweigh the wage losses of domestic workers.

The Political Economy Puzzle: Why is Capital Free but Labor Restricted?

If both labor and capital mobility create a net national welfare gain, why do governments **fiercely restrict** immigration while actively competing to attract FDI?

1. **Distributional Conflict & Voting Power**:
    - The losers from immigration (`native low-skilled workers`) are numerous, geographically concentrated, and have voting power. They fiercely oppose it.
    - The losers from capital outflow (`domestic capital owners`) are wealthy and influential, and they `lobby` the government to `allow` capital to leave (or offer tax breaks to keep it). The winners from capital inflow (labor) are diffuse and rarely organize to demand more FDI.

2. **Fiscal Burden:** Immigrants may use public services (schools, healthcare, infrastructure). If they are low-skilled, the short-term fiscal cost to the host government can be perceived as a burden, unlike capital, which pays corporate taxes.
3. **Cultural and Social Friction:** Labor mobility involves people, bringing cultural, linguistic, and social changes that trigger political backlash. Capital is impersonal and faces no such resistance.
4. **Sovereignty and Security:** Governments view control over their borders and population as a core element of national sovereignty. Capital flows are seen as purely economic.

Key Takeaways

1. **Factor mobility reallocates resources globally**, increasing total world output and efficiency.
2. **It creates sharp domestic winners and losers.** In the host country, the abundant factor gains and the scarce factor loses (and vice versa for the source country).
3. **Trade and Factor Mobility can be both substitutes and complements**, depending on whether the investment is horizontal (replacing trade) or vertical (creating global supply chains).
4. **Political reality diverges from economic theory:** Capital is highly mobile globally due to its political power and lack of social friction, while labor mobility remains heavily restricted due to distributional conflicts and national sovereignty concerns.

Mundell Model (1957) for Substitution

Mundell used the standard **Heckscher-Ohlin (H-O) framework**:

- **Two Countries:** Country A (`Capital-abundant`) and Country B (`Labor-abundant`).
- **Two Goods:** Good X (`Capital-intensive`) and Good Y (`Labor-intensive`).
- Initial State: There is free trade in goods , but `zero factor mobility`.
    - Because of their different factor endowments, Country A exports Good X, and Country B exports Good Y. Because capital is scarce in Country B, the return on capital (interest rate/profit) is higher in B than in A.

The Core Theorem: Perfect Substitution

This is the moment of Mundell's 1957 paper. As `capital moves from A to B`, the production capacities of both countries change.

- Country B, now flush with capital, starts producing more of the capital-intensive Good X `domestically`. It no longer needs to import Good X from Country A.
- Country A, having lost capital, shrinks its production of Good X and shifts toward Good Y.

- **The Conclusion (The Substitution Theorem):** The movement of capital **exactly offsets** the need to trade goods. Capital flows from the capital-abundant country to the labor-abundant country until the capital-labor ratios in both countries are identical.
- Once factor endowments are `equalized`, `advantage disappears`, and trade in goods drops to zero. Therefore, under Mundell's strict assumptions, trade in goods and international factor mobility are **PERFECT SUBSTITUTES**. You only need one or the other to achieve global efficiency.

Model Setup

- **Countries:** A (capital-abundant) and B (labor-abundant)
- **Goods:** X (capital-intensive) and Y (labor-intensive)
- **Factors:** K (capital) and L (labor)
- **Production Functions** (`identical technology` in both countries):
  - $X = F(K_X, L_X)$ where F is capital-intensive
  - $Y = G(K_Y, L_Y)$ where G is labor-intensive
- **Factor Endowments:**
  - Country A: (K_A, L_A) with $\frac{K_A}{L_A} > \frac{K_B}{L_B}$
  - Country B: (K_B, L_B)

Initial Equilibrium: Free Trade, No Factor Mobility

- Under the `Heckscher-Ohlin model` with free trade but `immobile factors`:
  - **Goods Prices** (equalized by free trade): $P_X^A = P_X^B = P_X$ $P_Y^A = P_Y^B = P_Y$
- Let the relative price be: $p = \frac{P_X}{P_Y}$
- **Factor Prices** (NOT equalized due to different endowments):
  - Using the `zero-profit conditions` under `perfect competition`:
    - $P_X = r \cdot a_{KX} + w \cdot a_{LX}$ and $P_Y = r \cdot a_{KY} + w \cdot a_{LY}$
      - Where: r = rental rate of capital (return on capital),
      - w= wage rate,
      - a_{ij} = unit input requirement of factor i in sector j

Key Result: Since Country A is capital-abundant, capital is relatively plentiful, so: $r^A < r^B$ (capital has higher return in B), $w^A > w^B$ (labor has higher wage in A).

- This is the Factor Price Equalization theorem does **NOT hold** when factors are immobile.
- **Trade Pattern:** Country A exports X (capital-intensive good), Country B exports Y (labor-intensive good). Trade volume: $T_0 > 0$

Introducing Factor Mobility: Capital Flows

Now allow capital to `move freely`. Capital seeks the highest return.

- **Arbitrage Condition:** Capital flows from A to B because $r^A < r^B$.
- Let K_F = amount of capital flowing from A to B.
- **New Factor Endowments:** Country A: $(K_A - K_F, L_A)$, Country B: $(K_B + K_F, L_B)$, Capital Flow Continues **Until**: $r^A = r^B = r$

The Substitution Theorem: Mathematical Proof

Step 1: Factor Price Equalization

- With `perfect factor mobility` and `identical technology`, capital flows until factor prices are equalized: $r^A = r^B = r^*$, $w^A = w^B = w^*$,
  - This is **Factor Price Equalization (FPE)**.

Step 2: Production Patterns Converge

- Since both countries now face `identical factor prices` (r^*, w^*) and have `identical technology`, they use the `same input coefficients`:

$$a_{KX}^A = a_{KX}^B = a_{KX}^*, \quad a_{LX}^A = a_{LX}^B = a_{LX}^*, \quad a_{KY}^A = a_{KY}^B = a_{KY}^*, \quad a_{LY}^A = a_{LY}^B = a_{LY}^*$$

Step 3: Capital-Labor Ratios Equalize

- The `capital-labor ratio` in each sector becomes identical:

$$\frac{K_X^A}{L_X^A} = \frac{K_X^B}{L_X^B} = \frac{a_{KX}^*}{a_{LX}^*} \frac{K_Y^A}{L_Y^A} = \frac{K_Y^B}{L_Y^B} = \frac{a_{KY}^*}{a_{LY}^*}$$

Step 4: Comparative Advantage Disappears

- Since both countries now have: `Identical` technology, `Identical` factor prices, `Identical` production techniques.
  - Their **opportunity costs** become `identical`: $MRT^A = MRT^B$,
  - Where MRT is the Marginal Rate of Transformation.

Step 5: Trade Volume Goes to Zero

- With `identical opportunity costs`, there is **no comparative advantage**, therefore: $T_1 = 0$.
- The amount of capital flow K_F that achieves this:
  - From the `full-employment conditions`:

$$K_X^A + K_Y^A = K_A - K_F, \quad L_X^A + L_Y^A = L_A, \quad K_X^B + K_Y^B = K_B + K_F, \quad L_X^B + L_Y^B = L_B$$

- Using the `fixed input coefficients`:

$$a_{KX}^* L_X^A + a_{KY}^* L_Y^A = K_A - K_F, \quad L_X^A + L_Y^A = L_A$$

- Solving for L_X^A and L_Y^A, then: $K_F = K_A - (a_{KX}^* L_X^A + a_{KY}^* L_Y^A)$.
  - This K_F is the exact amount of capital flow that equalizes factor prices and eliminates trade.

The Key Equation: Substitution Relationship

Before Factor Mobility:

- Trade volume: $T_0 > 0$,
- Capital flow: $K_F = 0$, `Factor price differential`: $\Delta r = r^B - r^A > 0$

After Factor Mobility:

- Trade volume: $T_1 = 0$, Capital flow: $K_F > 0$, `Factor price differential`: $\Delta r = 0$

The Substitution Theorem:

- $\boxed{T_0 + K_F \cdot \Delta r = \text{Constant}(\text{Global Efficiency})}$

Or more precisely: `Trade↓ ⇔ Factor Mobility↑` .

- They are **perfect substitutes** in achieving global efficiency.

Summary

Condition	Before Factor Mobility	After Factor Mobility
Factor Prices	$r^A \neq r^B, w^A \neq w^B$	$r^A = r^B = r^*, w^A = w^B = w^*$
Comparative Advantage	Exists ($MRT^A \neq MRT^B$)	None ($MRT^A = MRT^B$)
Trade Volume	$T_0 > 0$	$T_1 = 0$
Capital Flow	$K_F = 0$	$K_F > 0$

- **Conclusion:** `Factor Mobility≡Trade in Goods`.
- They are `mathematically equivalent` paths to the same efficient outcome.

Management and International Business View

Stephen Hymer (1960): Monopolistic Advantage Theory

- **The Core Question:** Why would a company go abroad at all? Foreign firms have a "home-field advantage" (they know the language, culture, laws, and customers). A foreign entrant suffers from a **"liability of foreignness."**
- **The Simple Answer:** A firm will only undertake the risk and cost of FDI if it possesses a **unique, monopolistic advantage** that local competitors do not have and cannot easily copy. This advantage compensates for the `liability of foreignness.`
- **Key Concept:** `Firm-Specific Advantages` (FSAs), such as proprietary technology, a world-famous brand, or superior management skills.
- **Simple Example:** Apple opens stores in China. Despite not knowing the local culture as well as Chinese firms, Apple succeeds because it has a `unique, globally desired product` and brand (its monopolistic advantage) that local firms cannot easily replicate.

Raymond Vernon (1966): Product Life Cycle (PLC) Theory

- **The Core Question:** *When* and *where* does a company decide to produce abroad?
- The Simple Answer: The location of production shifts as a `product matures`. It happens in three stages:
  1. **New Product:** Invented and produced in the home country (e.g., the US) to be close to R&D and wealthy early adopters. Exported to others.
  2. **Maturing Product:** Demand grows abroad. To save on shipping and tariffs, the firm builds factories in *other advanced countries* (e.g., Europe) to serve those local markets.
  3. **Standardized Product:** The product becomes a basic commodity. Competition is now all about price. The firm moves production to *developing countries* (e.g., Vietnam, Mexico) to take advantage of cheap labor, and exports the finished good back to the home country.
- **Simple Example:** The personal computer. Invented and made in the US (1970s). As demand grew, assembly moved to Europe and Japan (1980s). Today, PCs are standardized, so manufacturing is almost entirely in China and Southeast Asia for low-cost assembly.

Buckley & Casson (1976): Internalization Theory

- **The Core Question:** If a company has a `great technology advantages`, why doesn't it just *license* it to a local foreign company and collect a royalty fee? Why go through the hassle of building its own factory (FDI)?
- **The Simple Answer:** Because markets for knowledge and technology are **imperfect and risky**. If you license your secret technology to a foreign firm, they might steal it, produce low-quality goods that ruin your brand, or become a future competitor. Therefore, it is safer and more profitable to **"internalize"** the operation—meaning you keep it in-house by owning the foreign subsidiary yourself.
- **Key Concept:** Avoiding `market failure` in the transfer of knowledge.
- **Simple Example:** Coca-Cola has a secret syrup formula. It *could* license the recipe to an independent bottler in Brazil. But to prevent the recipe from being stolen or the quality from dropping, Coca-Cola uses FDI to own and strictly control its bottling operations (or tightly control joint ventures), keeping the process `internal`.

John Dunning (1980): The OLI Paradigm (Eclectic Paradigm)

- **The Core Question:** What is the *complete* checklist of conditions required for a firm to engage in FDI?
    - The Simple Answer: Dunning brilliantly combined the previous three theories into one master framework. He argued that a firm will only choose FDI if all three of the following conditions are met simultaneously:

1. **O - Ownership Advantage** (from Hymer): The firm must have a unique asset (brand, tech, patent) that gives it an edge over local rivals.
2. **L - Location Advantage** (from Vernon/Trade Theory): The foreign country must offer something attractive (cheap labor, large market, natural resources, or favorable taxes) that makes producing *there* better than producing at home.
3. **I - Internalization Advantage** (from Buckley & Casson): It must be more beneficial for the firm to own and control the foreign operation itself, rather than licensing or franchising it to a third party.

Summary

Scholar (Year)	Theory Name	The One-Sentence Summary
Hymer (1960)	Monopolistic Advantage	"You need a **special, unique advantage** to overcome the disadvantage of being a foreigner."
Vernon (1966)	Product Life Cycle	"Production moves from **rich countries** (innovation) to **poor countries** (cheap labor) as a product becomes standardized."
Buckley & Casson (1976)	Internalization Theory	"Keep it **in-house (FDI)** rather than licensing it, because licensing risks theft or quality loss."
Dunning (1980)	OLI Paradigm	"FDI only happens if you have **O**wnership advantages, **L**ocation advantages abroad, and **I**nternalization benefits."

Economic View

- Economists like **James Markusen (1983, 1984)** and **Elhanan Helpman (1984)** developed the **Complementarity Hypothesis** to explain this reality.
- They proved that under real-world conditions (imperfect competition, economies of scale, and fragmented production), factor mobility actually `creates` and `increases` international trade.

1. Vertical FDI and Global Value Chains (Helpman, 1984)

Elhanan Helpman shifted the focus from `where` capital goes to `how` production is organized.

- **The Logic:** In the real world, production is not a single step; it is a chain of stages (R&D →→ component manufacturing →→ assembly →→ marketing).
- **The Mechanism:** A Multinational Corporation (MNC) uses FDI to `fragment` its production process across borders to exploit different countries' `comparative advantages`. For example, a US tech firm might keep R&D in California (capital/skill-abundant) but use FDI to build an assembly plant in Vietnam (labor-abundant).
- **The Complementarity:** This capital movement **requires** the physical movement of goods. The US parent company must `export intermediate components`, `machinery`, and `blueprints` to the Vietnamese affiliate. The affiliate then exports the finished good back to the US or to third countries.
- **Result: Vertical FDI directly generates massive trade in intermediate goods.** Trade and FDI are complements because the global supply chain relies on both.

2. Economies of Scale and Intra-Firm Trade (Markusen, 1983, 1984)

James Markusen's `crucial contribution` was dropping Mundell's assumption of `constant returns to scale` and `perfect competition.` Markusen introduced **increasing returns to scale (economies of scale)** and **imperfect competition (oligopoly/monopolistic competition)**.

- **The Mundell Flaw:** `Mundell` assumed that if capital moves to Country B, Country B just produces the good locally, replacing imports. But `Markusen` pointed out that if a single plant in Country B is too small to achieve the minimum efficient scale, it will fail.
- **The Markusen Solution (Multi-Plant Firms):** To achieve economies of scale, an MNC uses FDI to build a network of plants `globally`. However, to keep costs down, the parent company `centralizes` the production of core, highly `specialized components` or capital equipment at the home plant.
- **The Mechanism (Intra-firm Trade):** The parent company then uses **FDI** to establish foreign distribution or assembly affiliates, and uses **trade** to continuously ship those specialized core components from the home plant to the foreign affiliates.
- **Result:** FDI does not replace trade; it **locks in** a continuous flow of `intra-firm trade`. The larger the FDI network, the more intermediate parts the parent company exports to its own subsidiaries.

3. The "Knowledge-Capital" Model (Markusen, 1995; Carr, Markusen, et al., 2001)

Markusen later synthesized his work into the **Knowledge-Capital (KK) model**, which mathematically unified both `Horizontal FDI` (duplicating the same production abroad) and `Vertical FDI` (fragmenting production).

The KK model proves that trade and FDI are complements based on the specific characteristics of the two countries:

1. **Similarity in Size and Wealth:** If Country A and Country B are `similar` in `size and wealth`, firms engage in **Horizontal FDI** to avoid trade costs (tariffs/transport) and be close to consumers. Even here, trade and FDI coexist because the MNC still trades specialized services and headquarters' knowledge.
2. **Differences in Factor Endowments:** If Country A is skill-abundant and Country B is unskilled-labor-abundant, firms engage in **Vertical FDI**. As proven by Helpman, this heavily *complements* trade in intermediate goods.

The Mathematical Intuition of Complementarity:

- Let T be Trade and K be Capital Flow (FDI).
- Mundell: $\frac{\partial T}{\partial K} < 0$ (An increase in FDI decreases Trade).
- Markusen/Helpman: $\frac{\partial T}{\partial K} > 0$ (An increase in FDI *increases* Trade, specifically in intermediate goods and intra-firm trade).

Helpman's Vertical FDI Model (1984)

Before Helpman, trade theory treated firms as **black boxes** — they just turned inputs into outputs. Helpman opened the box and modeled the **internal structure of the multinational firm (MNF)**:

1. **Firms possess "headquarters services" (H)** — knowledge, management, brand, R&D — that are **non-rival** (can be used in multiple plants simultaneously at zero marginal cost).
2. **Setting up headquarters is costly** (fixed cost), but once established, it can serve plants in multiple countries.
3. **Production can be fragmented**: Different stages can be located in different countries to exploit factor-price differences.

This is what makes FDI and trade **complements**: the headquarters sends knowledge/capital (FDI) to foreign plants, and those plants ship `intermediate goods back (Trade)`.

Model Setup

- **Two countries**: Home (H) and Foreign (F), **Two factors**: Capital (K) and Labor (L)
- Two goods:
  - Y = a `homogeneous` numeraire good (produced under perfect competition, constant returns)
  - X = a `differentiated` good (produced by firms with market power)
- **Factor endowments**: Country H is capital-abundant, Country F is labor-abundant
  - $\frac{K_H}{L_H} > \frac{K_F}{L_F}$

The Firm's Technology

A firm producing good X uses `three inputs`:

- **Headquarters services (H)** — a firm-specific, non-rival input,
- **Capital (K)** — used in production plants, **Labor (L)** — used in production plants
- **Production function at a single plant i:** $x_i = f(H, K_i, L_i)$
  - **Key property**: H is a **public good within the firm**.
    - If the firm has two plants (one in H, one in F), the `same` H serves both:
    - $x_H = f(H, K_H, L_H), x_F = f(H, K_F, L_F)$.
    - `Total output` of the firm: $X = x_H + x_F$.

Cost Structure

The firm faces two types of fixed costs:

- F_p = `fixed cost` of setting up a **production plant**, F_h = `fixed cost` of setting up **headquarters**
- Total cost for **a national firm** (one plant in Home only):
  - $C_N = F_h + F_p + c(w_H, r_H) \cdot x_H$
- Total cost for **a multinational firm** (headquarter and plant in Home, plant in Foreign):
  - $C_M = F_h + 2F_p + c(w_H, r_H) \cdot x_H + c(w_F, r_F) \cdot x_F + \tau \cdot T$

Where:

- τ = `trade cost (iceberg costs)` for shipping intermediates between plants,
- T = volume of intra-firm trade

The Firm's Optimization Problem

Step 1: Choose Firm Type

The firm chooses between:

- **National (N)**: One plant at home, serves both markets via `exports`
  - Profit as a national firm: $\pi_N = P_H \cdot x_H + P_F \cdot x_H^{export} - C_N - \tau \cdot x_H^{export}$
- **Multinational (M)**: Headquarters and plant at home, another production plant in foreign country
  - Profit as a multinational: $\pi_M = P_H \cdot x_H + P_F \cdot x_F - C_M$

Step 2: Choose Location of Production Stages (Vertical FDI)

If the firm becomes `multinational`, it fragments production:

- **Headquarters-intensive stage** (R&D, management) → stays in Home (capital/skill-abundant)
- **Labor-intensive stage** (assembly, manufacturing) → moves to Foreign (labor-abundant)

Why? To minimize costs by `exploiting factor-price differences`:

- $w_F < w_H$ (lower wages in Foreign),
- $r_H < r_F$ (lower capital costs in Home)

Step 3: Optimal Allocation

The firm `minimizes total cost` subject to producing total output X:

$$\min_{x_H, x_F} \quad c(w_H, r_H) \cdot x_H + c(w_F, r_F) \cdot x_F + \tau \cdot T, \quad \text{Subject to} : x_H + x_F = X$$

- The trade volume T is not a constant — it depends on how much the foreign plant produces $T = T(x_F)$. Because the foreign plant needs intermediate goods (components, headquarters services) shipped from the parent company. The more the foreign plant produces ($x_F \uparrow$), the more intermediates it needs, so $T \uparrow$
  - From $x_H + x_F = X$, we get: $x_H = X - x_F$.
  - Take the derivative with respect to x_F and set it to zero
    - $\frac{d}{dx_F}[c_H \cdot X - c_H \cdot x_F + c_F \cdot x_F + \tau \cdot T(x_F)] = 0$
- **First-order condition:** $c(w_H, r_H) = c(w_F, r_F) + \tau \cdot \frac{\partial T}{\partial x_F}$
- **Interpretation**: The firm allocates `production between Home and Foreign` until the marginal cost (including trade costs) is equalized across plants.

The Complementarity Result

Intra-Firm Trade Volume

- When the firm operates as a multinational, it must `ship intermediate goods` between the headquarters and the foreign plant.
- Let's denote:

$$T = \text{intra-firm trade} = \text{intermediates shipped from H to F}$$

Key insight:

- The foreign plant's output depends on the headquarters services it receives:
  - $x_F = g(H, K_F, L_F)$.
- And the headquarters must ship components/knowledge to support that plant:
  - $T = \phi \cdot x_F$, Where $\phi > 0$ is the **trade intensity** of foreign production.

The Complementarity Derivative

Total trade between the two countries includes:

1. **Intra-firm trade** (T): Intermediates between `parent` and `foreign affiliate`.
2. **Arm's-length trade**: other goods

- Focusing on the FDI-generated trade:
  - $T_{FDI} = \phi \cdot x_F(K_F)$, where K_F is the capital invested in the foreign plant (FDI).
  - **The key derivative:** $\frac{\partial T_{FDI}}{\partial K_F} = \phi \cdot \frac{\partial x_F}{\partial K_F} > 0$
- Since $\frac{\partial x_F}{\partial K_F} > 0$ (more capital → more output at the foreign plant), we get:
  - $\boxed{\frac{\partial T_{FDI}}{\partial K_F} > 0}$

This is the Complementarity Theorem:

- An increase in FDI (K_F) **increases** trade (T).

Does This Happen? (Intuition)

Let me walk through the logic step-by-step:

1. **Factor-price difference exists**: $w_F < w_H$
2. **Firm responds**: Sets up a labor-intensive plant in Foreign (FDI = K_F)
3. **Plant needs inputs**: The foreign plant needs headquarters services, components, and machinery from the parent
4. **Trade is born**: The parent ships these inputs to the foreign plant (Trade = T)
5. **Feedback loop**: The more the firm invests abroad (higher K_F), the more inputs it must ship (higher T)

Contrast with Mundell:

- **Mundell**: Capital moves → production moves entirely → no need to trade
- **Helpman**: Capital moves → production *fragments* → must trade intermediates to connect the fragments

Equilibrium Conditions

In general equilibrium, Helpman derives several key results:

Result 1: FDI Happens When Countries Differ in Factor Endowments

- A firm becomes `multinational` ($K_F > 0$)
  - if: $\pi_M > \pi_N \Rightarrow (w_H - w_F) \cdot L_F > F_p + \tau \cdot T$
- **Interpretation**: The `wage savings` from locating labor-intensive production in the low-wage country must exceed the fixed cost of the foreign plant plus trade costs.
- **Implication**: Vertical FDI is more likely when:
  - Factor endowment differences are **large** ($w_H \gg w_F$)
  - Fixed costs of foreign plants are **small** (F_p low)
  - Trade costs are **low** (τ small)

Result 2: FDI Increases Trade in Intermediates

- In equilibrium: $T^* = \phi \cdot x_F^*(K_F^*)$, where K_F^* is the equilibrium FDI.
- As factor endowment differences grow:
  - K_F^* increases (more FDI)
  - x_F^* increases (more foreign production)
  - T^* increases (more intra-firm trade)

Result 3: FDI Affects Factor Prices

- FDI flows from capital-abundant Home to labor-abundant Foreign:
  - **In Home**: Capital becomes slightly scarcer → r_H rises, w_H falls
  - **In Foreign**: Capital becomes more abundant → r_F falls, w_F rises

This **converges** factor prices across countries, but does **not** eliminate trade (unlike Mundell), because the trade is now in intermediates, not final goods.

Summary

Helpman's Key Equations

1. **Firm's profit as multinational:**
   - $\pi_M = P_H x_H + P_F x_F - F_h - 2F_p - c_H x_H - c_F x_F - \tau T$
2. **Intra-firm trade:** $T = \phi \cdot x_F$
3. **Complementarity condition:** $\frac{\partial T}{\partial K_F} = \phi \cdot \frac{\partial x_F}{\partial K_F} > 0$
4. **FDI threshold:** $K_F > 0 \iff (w_H - w_F)L_F > F_p + \tau\phi x_F$
5. **Equilibrium trade volume:** $T^* = \phi \cdot g(H, K_F^*, L_F)$

Empirical Implications

`Helpman's model` makes several testable predictions that have been confirmed by data:

1. **Vertical FDI is larger between countries with different factor endowments** (e.g., US-Mexico, Germany-Poland)
2. **Intra-firm trade is a large share of total trade** (about 30% of US trade is `intra-firm`)
3. **FDI and trade are positively correlated** across country pairs
4. **Lower trade costs increase both FDI and trade** (they are complements, not substitutes)

`Helpman's 1984 model` revolutionized international economics by showing that

- the multinational firm is not just a vehicle for moving capital — it is a network that connects production stages across borders through trade.
- The mathematical essence is simple but profound:
  - FDI creates foreign plants→foreign plants need inputs →inputs must be traded. so $\boxed{\frac{\partial T_{FDI}}{\partial K_F} > 0}$

This is why, in the real world, we see **massive FDI and massive trade happening simultaneously** — they are two sides of the same global production network.